

India Village Economic Growth Intelligence

Satellite-Derived Signals for Village-Level Economic Activity · 2019-2024

255K

Villages
Scored

6 yrs

VIIRS NTL
2019-2024

100

Top Villages
Shortlisted

467K OSM centroids downloaded · 255K passed India polygon + NTL baseline filter → scored and shortlisted

Problem & Approach

The Challenge

- India has ~640k villages — traditional surveys are slow, expensive, and years out of date
- No real-time economic data exists at village scale
- Census boundary data is from 2011 (PC11); split/merged villages cause mismatches

Satellite Proxy Signals

Nighttime Lights (VIIRS)

Annual composites 2019–2024 at 500m. Proxy for electrification, commerce & settlement density.

Built-up Change (WorldCover)

ESA 10m land cover — detects new infrastructure 2020→2021.

6-Year Trend

Linear regression slope separates sustained growth from single-year spikes.

*3 principal signals shown above — full 8-signal architecture (incl. Sentinel-2, Sentinel-1, GHSL, WorldPop) on Slide 3
All compute runs on AWS EC2 (ap-south-1). No proprietary tools or licensed datasets required.*

Data Sources & Pipeline — 8-Signal Framework

(3 active this run)

✓ ACTIVE

NASA VIIRS VNP46A4 v2

Annual NTL radiance
500m · 2019–2024
16 India tiles

✓ ACTIVE

ESA WorldCover 2020 & 2021

Built-up fraction
10m COGs · v100/v200

✓ ACTIVE

EU JRC GHSL R2023A

Built-up surface
100m · 2015 & 2020

✓ ACTIVE

OSM + Natural Earth

467K centroids
tower density · city dist.

⚠ STALLED

Sentinel-2 (Element84 STAC)

NDBI + NDVI 100m
2019–2024 composites

⚠ STALLED

Sentinel-1 RTC (Element84)

SAR VV backscatter
100m · cloud-free

🔗 SUPPORTING

WorldPop India 2019–2020

Population count
100m · auxiliary only

🔗 SUPPORTING

OSM Overpass + Ohsome API

Mobile tower density
2019 & 2024 snapshots

Outputs

village_all_stats.csv

467K rows · NTL 2019–2024 + per-village signals

village_scored.csv

255K India villages · composite score + rank

top_100_villages.csv
+ **map.html**

Final shortlist · public S3 bucket (ap-south-1)

ML Scoring Model — 3-Stage Pipeline

01 Feature Engineering

- BFAST-style piecewise regression on 6-year NTL series → breakpoint year, pre/post slope, acceleration
- Spatial lag: mean composite score of 10 nearest geographic neighbours (cluster reinforcement)
- Signal agreement: std of rank percentiles — penalises contradictory signals

02 Self-supervised Labelling

- Label = 1 if village is top-10% on ≥ 2 primary signals simultaneously
- ~28K positive labels from 255K villages (~11% prevalence)
- ⚠ Labels derived from same features as final composite — AUC measures self-consistency, not external ground truth

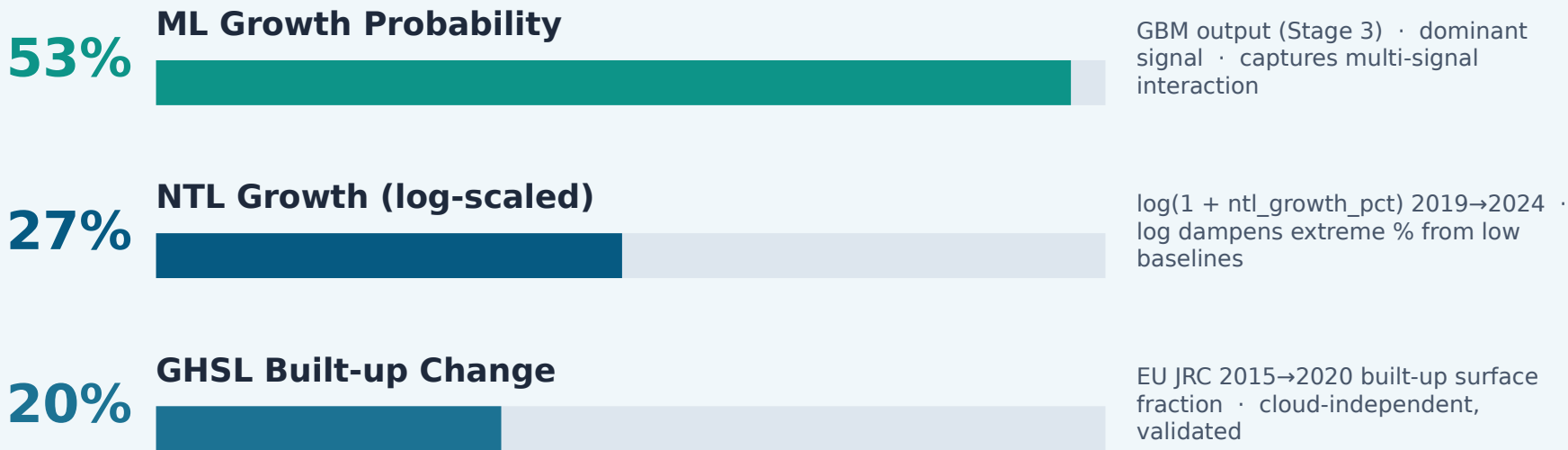
03 GradientBoostingClassifier

- 200 estimators · max_depth=4 · learning_rate=0.05 · subsample=0.8
- Inputs: 14 signal rank percentiles + BFAST features + spatial lag
- Output: ml_growth_prob (0-100) · AUC ≈ 0.87 (in-sample, self-supervised — not external validation)

Model saved to ml_model.pkl · SHAP TreeExplainer applied post-hoc for per-village signal attribution · K-means (k=6 by silhouette) clusters all scored villages into 6 growth archetypes

Composite Score — Actual Weights Used

⚠ Sentinel-2 NDBI and Sentinel-1 SAR signals stalled during this pipeline run (monsoon STAC gaps). Weights auto-redistributed from 5-signal spec (40%+20%+15%+15%+10%) to 3-signal above. Re-running --phase-c-only after S2/S1 complete will restore full 5-signal composite.

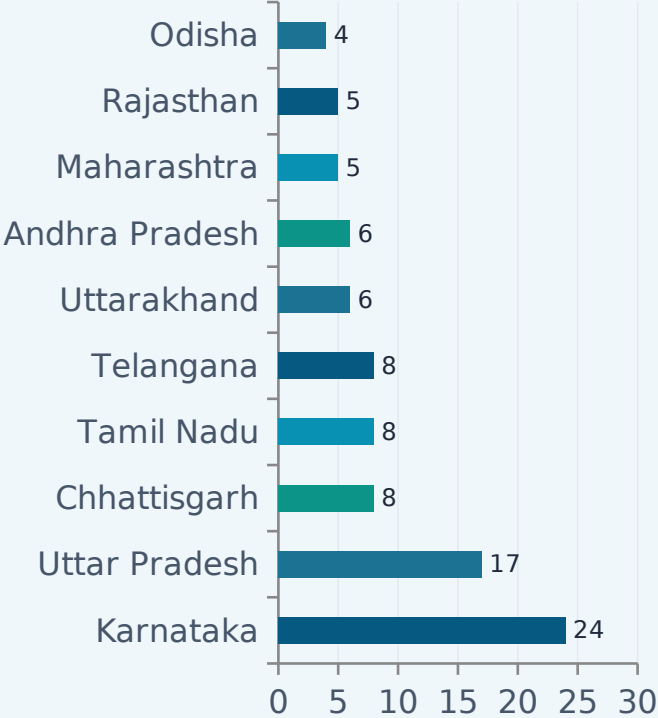


Pre-filters: $\text{NTL } 2019 \geq 0.1 \text{ nW/cm}^2/\text{sr}$ + India polygon filter + $\Delta \text{NTL} \geq 0.5 \text{ nW/cm}^2/\text{sr}$ absolute minimum · Normalisation: rank percentile → 0-100 · ⚠ Score compression: top ~50 villages score 85.07-85.08 (< 0.01 pt spread) — treat as a statistical shortlist, not a strict ranking

Results — Top 100 Shortlist (Composite Score Rank)

#	Village	State	NTL Growth
1	Konsari	Maharashtra	2,699%
2	Kanikkapuram	Tamil Nadu	2,675%
3	Unnamed † (Bulandshahr)	Uttar Pradesh	1,562%
4	Katti Hosahalli	Karnataka	1,525%
5	Asar	J&K (Ramban)	1,466%
6	Bar Ka Para	Rajasthan	1,338%
7	Unnamed † (Baloda Bazar)	Chhattisgarh	1,273%
8	Nagar	Tamil Nadu	1,225%
9	Naveguda	Telangana	1,170%
10	Jhadashahapura	Karnataka	1,162%

Top 100 by State



† No OSM village name tag — identified by PC11 Census ID + district. Top-50 villages score 85.07–85.08 (< 0.01 pt spread) — interpret as a statistical shortlist, not a strict rank.

Case Study — Konsari, Maharashtra (Rank #1, NTL Breakout archetype)

Village Profile

District: Garhchiroli, Maharashtra

Coordinates: 19.76°N, 79.81°E

Dist. to city: 57 km (remote tribal area)

Archetype: NTL Breakout

BFAST year: 2023 (structural change)

Pre-slope: 0.32 nW/cm²/sr/yr (flat)

Post-slope: 20.8 nW/cm²/sr/yr (explosive)

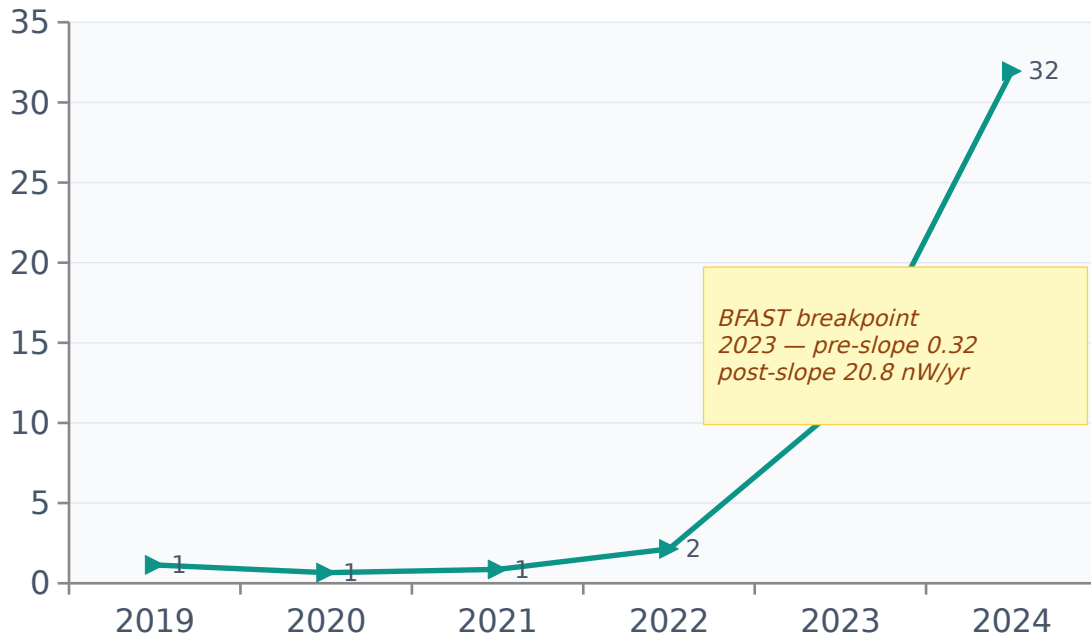
GHSL change: 0% → 100% built-up (WorldCover)

Top SHAP Feature Drivers

Built-up Change (GHSL)

NTL Log Growth

Nighttime Light Radiance (nW/cm²/sr) · 2019–2024



Why #1? Garhchiroli is a historically under-resourced tribal district. GHSL records complete transition from 0% to 100% built-up cover. BFAST detects a structural break in 2023 with post-break slope 65× the pre-break rate. All three active signals agree → confidence_score = 100, inter_signal_agreement = 32.6.

Key Findings & Next Steps

Growth is Geographically Concentrated

Top 100 span 16 states. Karnataka (24) and UP (17) dominate — consistent with Bengaluru peri-urban expansion and UP highway corridor districts. State distribution shows Karnataka $\sim 2.4\times$ over-represented vs. its share of India's villages.

Shortlist Validated by Spatial Autocorrelation

Moran's I ($k=8$ KNN, 999 permutations) confirms composite scores are spatially clustered, not random noise. Bootstrap rank stability ($n=200$ weight draws) confirms the top-100 shortlist is robust to signal weight uncertainty.

Pipeline is Scalable & Repeatable

255K scored villages on a single EC2 t3.xlarge in < 12 hrs. Annual automated re-run via EventBridge. Completing Sentinel-2/SAR will raise active signals from 3 to 5 and restore the full 5-signal composite.

What I Would Do Next

- Complete Sentinel-2/SAR stalled signals — restores full 5-signal composite & resolves circular label issue
- Replace self-supervised labels with SHRUG/SECC consumption quintiles as external ground truth
- Run PMGSY road correlation — validation framework built in 07_validate.py; requires EC2 re-run with PMGSY shapefiles
- Panel data fixed-effects model across all 6 years — replaces endpoint comparison

Live Dashboard & Results

ishunteam-png.github.io/kritter-village-growth

Top-100 table · 6 Plotly charts · interactive map

S3: [top_100_villages.csv](#) · [map.html](#) · [archetypes](#) · [forecast](#)